Year 8 Science – Booklet 1: Chemistry

Chemical Reactions (Elements, Compounds and Reactions) — ANSWERS

Quick Test (p.45)

1. CO■ and H■O (O■ is an element, not a compound – it's made of only one type of atom).
2. The number of protons in the nucleus of an atom of that element.
3. 2.
4. 3 (carbon, hydrogen and oxygen).

Quick Test (p.47)

1. nitrogen + oxygen → nitrogen oxide
2. Potassium sulfate.
3. Any three of: conducts electricity/heat, ductile, malleable, shiny, sonorous, high melting/boiling point.
4. Any three of: poor conductor of electricity/heat, often low melting/boiling point, often a gas at room temperature, often lower density than metals, dull, brittle (if solid).

Vocabulary Builder (18 marks)

1. Match the reaction type to its description (4 marks)

Oxidation → oxygen is added to a substance
Electrolysis → using electricity to break down a substance into simpler substances
Thermal decomposition → using heat to break down a substance into simpler substances
Combustion → when a fuel combines with oxygen
Reduction → oxygen is removed from a substance

2. Complete the sentences about metal extraction (5 marks)

Some metals like gold are not reactive and are found as **native** elements. But most metals are reactive and are found in a **compound** trapped inside rocks. **Ores** are rocks that contain enough metal to make it worthwhile extracting the **pure** metal from them. Ores undergo **reduction** reactions to extract the metal from its compound.

3. Methane combustion: methane + oxygen → carbon dioxide + water

a). Methane and oxygen.

b). Carbon dioxide and water.

c). Methane.

d). Oxygen (the only element among the four substances named – the others are all compounds).

e). Methane, carbon dioxide, water.

f). Methane.

4. Match the metal property to its definition (3 marks)

Conductor → allows heat and electricity to easily pass through
Sonorous → makes a ringing noise when hit
Ductile → can be drawn into wires
Malleable → bends easily

Maths Skills (13 marks)

1. Halogen melting points bar chart

a). x-axis: Halogen element. y-axis: Melting point (°C).

b). Bromine.

c). Melting point increases as you go down group 7 (from fluorine, the lowest, to iodine, the highest).

2. Earth's crust pie chart

a). Oxygen.

b). Approximately 28% (reading the silicon slice of the chart).

c). Aluminium (the most abundant metal in the Earth's crust, after the non-metal/metalloid oxygen and silicon).

3. Malachite (58% copper, costs £244/kg, copper sells for £16)

a). mass of copper = 58% × 1000 g = **580 g**

b). No – even though malachite contains a high percentage of copper (58%), the cost of buying/processing 1 kg (£244) is far greater than the value of the copper that can be extracted from it (£16). Since it would not be economically worthwhile to extract the copper, this sample fails the definition of an ore (a rock where extracting the metal is worthwhile).

Testing Understanding (35 marks)

1. Periodic table extract

a). Li and K (both Group 1 – also accept F and Cl, Group 7).

b). K and Fe (both in the same period – also accept other same-row pairs shown).

c). Li (or K).

d). F (or Cl).

e). He.

2. Formula table (8 marks)

Name	Formula	No. elements	Total atoms
Water	H₂O	2	3
Sulfur dioxide	SO₂	2	3
Lead oxide	PbO	2	2
Calcium chloride	CaCl₂	2	3

3. Ethanol fuel in Brazil

a). Combustion (oxidation).

b). ethanol + oxygen → carbon dioxide + water

c). Bubble the gas produced through limewater – if carbon dioxide is present, the limewater turns cloudy/milky.

4. Metals vs non-metals (6 marks)

On the left/centre of the table → Metals
Solids at room temperature → Metals
Gases at room temperature → Non-metals
Malleable → Metals
Oxides are acidic → Non-metals
Oxides are basic → Metals

5. Copper's properties and uses

a). Cooking pans: good conductor of heat, so heat spreads evenly through the pan.

b). Electrical wires: good conductor of electricity, and ductile (can be drawn into wires).

c). Plumbing pipes: malleable (easy to shape into pipes) and resistant to corrosion by water.

6. Ros: 2.3 g magnesium + 1.6 g oxygen → magnesium oxide

a). Oxidation.

b). magnesium + oxygen → magnesium oxide

c). By conservation of mass: 2.3 + 1.6 = **3.9 g**

Working Scientifically (28 marks)

1. Derek's conductivity experiment

a). The material being tested.

b). The bulb would light up (glow) when that material completes the circuit, showing current is flowing through it.

c). Use crocodile clips to hold/connect the material firmly in the circuit.

d). The wires or material could become hot, and there is a small risk of electric shock – care should be taken with exposed wires and connections.

e). As a metalloid, silicon would be expected to conduct electricity, but not as well as a true metal (metalloids have intermediate conductivity between metals and non-metals) – so the bulb might glow only dimly, or not light at all if the current is too weak.

2. Kabir's calcium + water experiment

a). Conical flask.

b). (Electronic/digital) mass balance.

c). Time (as the reaction proceeds).

d). The mass reading will decrease over time, because the reaction produces hydrogen gas which escapes from the open flask into the air – since this gas has mass and leaves the system, the total mass on the balance falls.

3. Lithium, sodium and potassium stored in oil

a). These metals are highly reactive and react vigorously with water and with oxygen in the air, so storing them under oil keeps them away from air and water, preventing them reacting/corroding.

b). Wear eye protection; use tweezers/forceps (not bare hands) to handle the metals; use a safety screen and stand well back when adding them to water, as the reactions can be vigorous.

c). It tells us that all three metals are less dense than water (an object floats if it is less dense than the liquid it's placed in).

d). The indicator would turn purple/blue – the reaction produces an alkaline metal hydroxide solution (high pH), and universal indicator turns purple/blue in strongly alkaline conditions.

4. Mendeleev's periodic table

a). By increasing relative atomic mass (atomic weight).

b). He left gaps for undiscovered elements and used them to successfully predict the existence and properties of elements not yet found, unlike earlier scientists such as Dalton who simply listed known elements.

c). The modern table is arranged by increasing atomic number (number of protons), not atomic mass, and includes many more elements discovered/synthesised since Mendeleev's time (e.g. the noble gases and further transition metals).

5. Nikita's thermal decomposition of calcium carbonate (5 marks)

1. Heat the calcium carbonate strongly on a gauze and tripod over a Bunsen burner (on a heat-resistant mat) until it thermally decomposes into calcium oxide and carbon dioxide gas (which escapes). 2. Allow

the calcium oxide to cool. 3. Add water to the calcium oxide – it reacts to form calcium hydroxide, which partly dissolves. 4. Filter the mixture through filter paper in a filter funnel to remove any undissolved solid. 5. The clear filtrate collected is limewater (a solution of calcium hydroxide in water).

Science in Use (12 marks)

1. Structures of carbon

a). Diamond.

b). Diamond and graphene (2 marks).

c). C.

d). Group 4.

e). Period 2.

f). 6.

2. Discovery of metals (bronze = copper + tin)

a). Sn.

b). In the transition metals block, in the middle of the periodic table.

c). Green (or blue-green).

d). Reduction (using carbon to reduce the metal compound and release the pure metal).

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